



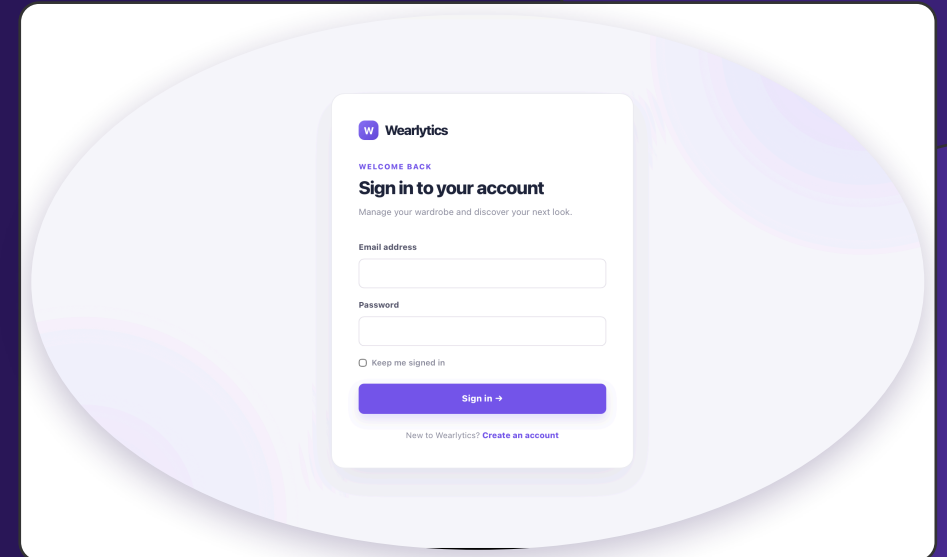
AI-Powered Wardrobe & Outfit Recommendation System

MCA MINI PROJECT — SECOND INTERIM PRESENTATION

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01 · PROJECT INTRODUCTION

What is Wearlytics?

Wearlytics is a web-based smart wardrobe platform that helps users organize clothing, analyze wardrobe usage and receive personalized outfit recommendations.

1 Project Overview

- Digital wardrobe for personal clothing
- Upload and organize clothing items
- AI-assisted clothing classification and color detection
- Personalized outfit recommendations from available items

2 Problem Statement

- Managing a large wardrobe can be difficult
- Hard to remember available clothing and previous usage
- Choosing suitable outfit combinations takes time
- Repeated use of the same items is difficult to track

3 Objectives

- Digitize and organize the personal wardrobe
- Assist clothing classification and color identification
- Generate personalized outfit recommendations
- Support planning, wear tracking and wardrobe analytics

02 · LITERATURE SURVEY

Review of Relevant Research

No.	Research Paper	Author	Year	Publisher	Methods / Technologies	Limitations
01	Hybrid-Hierarchical Fashion Graph Attention Network for Compatibility-Oriented and Personalized Outfit Recommendation	Sajjad Saed & Babak Teimourpour	2026	Elsevier	GNN, GAT, hierarchical graph, multimodal learning, attention	Static data, computational complexity, category imbalance; higher-order relations not fully explored.
02	OutfitAI: Shop the Outfit with a Deep Learning-based Intelligent Expert System	Emanuele Balloni et al.	2025	Springer	Semantic Segmentation, CNNs, Transformers, Transfer Learning, Triplet Hard Loss	High computational and memory requirements; possible deployment latency.
03	Using Large Multimodal Models to Predict Outfit Compatibility	Chia-Ling Chang, Yen-Liang Chen & Dao-Xuan Jiang	2025	Elsevier	Gemini VQA, keyword extraction, BEiT-3, multimodal fusion, BPR	Mainly top-bottom compatibility; not an integrated personal wardrobe system.
04	Enhanced Content-Based Fashion Recommendation System through Deep Ensemble Classifier with Transfer Learning	Suvarna Buradagunta & Sivadi Balakrishna	2024	Springer	Transfer Learning, CNNs, Deep Ensemble, Cosine Similarity	Content-based similarity; dataset diversity and generalization can be improved.
05	Unifying Heterogeneous and Homogeneous Relations for Personalized Compatibility Modeling	Jinhuan Liu, Lei Hou, Xu Yu, Xuemeng Song & Zhaochun Ren	2024	Elsevier	Graph Neural Networks, relation modelling, multimodal features, adaptive neighbour sampling	Comprehensive relation modelling, sampling and multimodal representation remain challenging.

RESEARCH GAP

“Existing research focuses mainly on clothing classification, outfit compatibility and personalized recommendation, with limitations in practical wardrobe context and integrated user-level management. Wearlytics combines AI clothing analysis, personalized outfit recommendation, laundry awareness, feedback learning, outfit planning and wardrobe analytics in a unified system.”

03 · DRAWBACKS OF THE EXISTING SYSTEM

Existing System / Approach	Drawback
Manual wardrobe management	Organizing and managing a large clothing collection manually requires effort.
Manual clothing identification	Users may need to manually enter clothing category and details.
Manual outfit selection	Finding suitable combinations can be time-consuming.
Generic outfit suggestions	User's own wardrobe and preferences may not be fully considered.
Clothes availability not always considered	Clothes in laundry may be included in practical outfit selection.
Separate wardrobe activities	Planning, wear tracking and analytics may not be integrated in one workflow.

KEY PROBLEM

Existing approaches often require manual management and outfit selection, with limited integration of AI analysis, personalization, availability, planning and wardrobe usage insights.

04 · PROPOSED SYSTEM

Overview

Wearlytics allows users to build a digital wardrobe by uploading clothing images. AI assists with clothing classification and dominant-color detection. A weighted recommendation engine generates suitable outfit combinations using color, category, occasion, season, preferences, novelty and favorites. Users can also manage laundry status, save outfits, plan weekly looks, track clothing usage and view wardrobe analytics.

How it addresses existing limitations:

- AI analysis reduces manual clothing categorization
- Personalized scoring improves outfit selection
- Laundry-aware recommendations avoid unavailable items
- Planning, wear tracking and analytics are integrated

Advantages of the Proposed System

- AI-assisted clothing classification
- Color-aware outfit matching
- Personalized recommendations using feedback and favorites
- Laundry-aware outfit selection
- Seven-day weekly outfit planning
- Wear tracking and wardrobe utilization insights
- Integrated wardrobe, recommendation and analytics workflow

05 · TECHNOLOGIES USED

Technology Stack

Technologies and tools used to develop the Wearlytics web application

FRONT-END

HTML5 · Jinja2 · CSS3 · Bootstrap 5.3
JavaScript · Google Fonts & Icons

Responsive UI, templates and client-side interactions

BACK-END

Python · Flask · Flask-SQLAlchemy
Flask-Login · Werkzeug

Application logic, authentication and secure file handling

DATABASE

MySQL · SQLite (development) · SQLAlchemy
PyMySQL

Relational storage for users, wardrobe, outfits and plans

DEVELOPMENT / FRAMEWORKS

Flask Application Factory · Flask-Migrate / Alembic
python-dotenv · Git / GitHub · Pytest

Modular architecture, migrations, environment and testing

AI / MACHINE LEARNING

PyTorch · Torchvision · MobileNetV2
Scikit-learn · NumPy · Pillow

Clothing classification and dominant-color detection

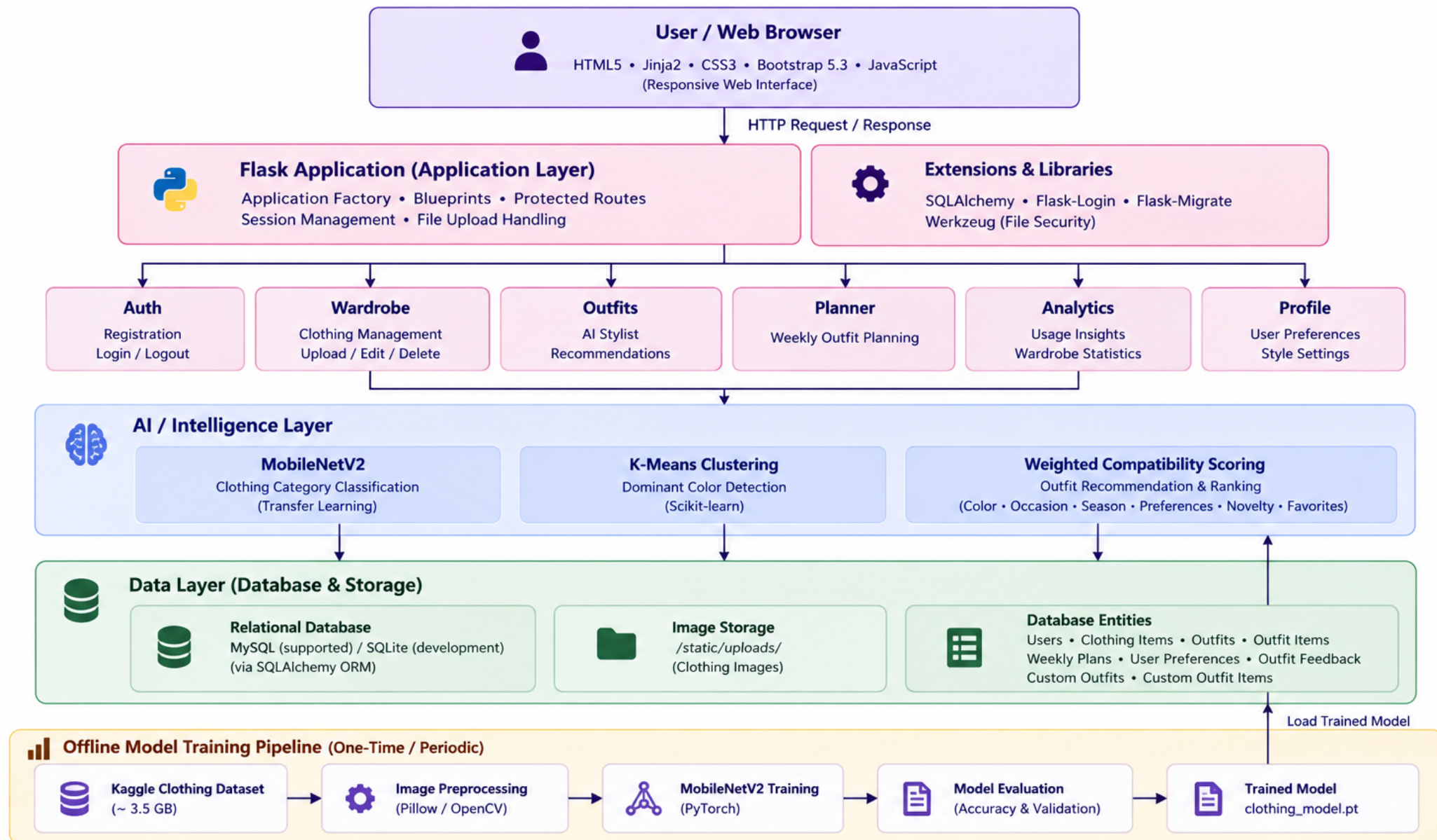
COMPUTER VISION

OpenCV · Image preprocessing
K-Means clustering · Dominant-color detection

Image processing and color analysis

Core stack: Python + Flask + SQLAlchemy | AI: MobileNetV2 + K-Means | Database: MySQL / SQLite

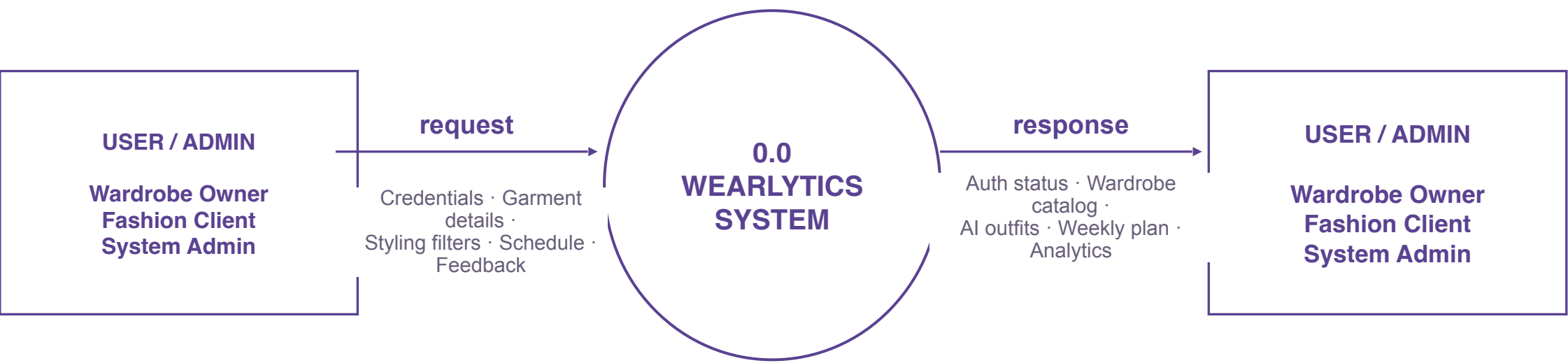
07 · SYSTEM ARCHITECTURE



06 · SYSTEM DESIGN — DFD LEVEL 0

(User → Wearlytics System → User)

Highest-level view: boundary interactions between actors and the Wearlytics system

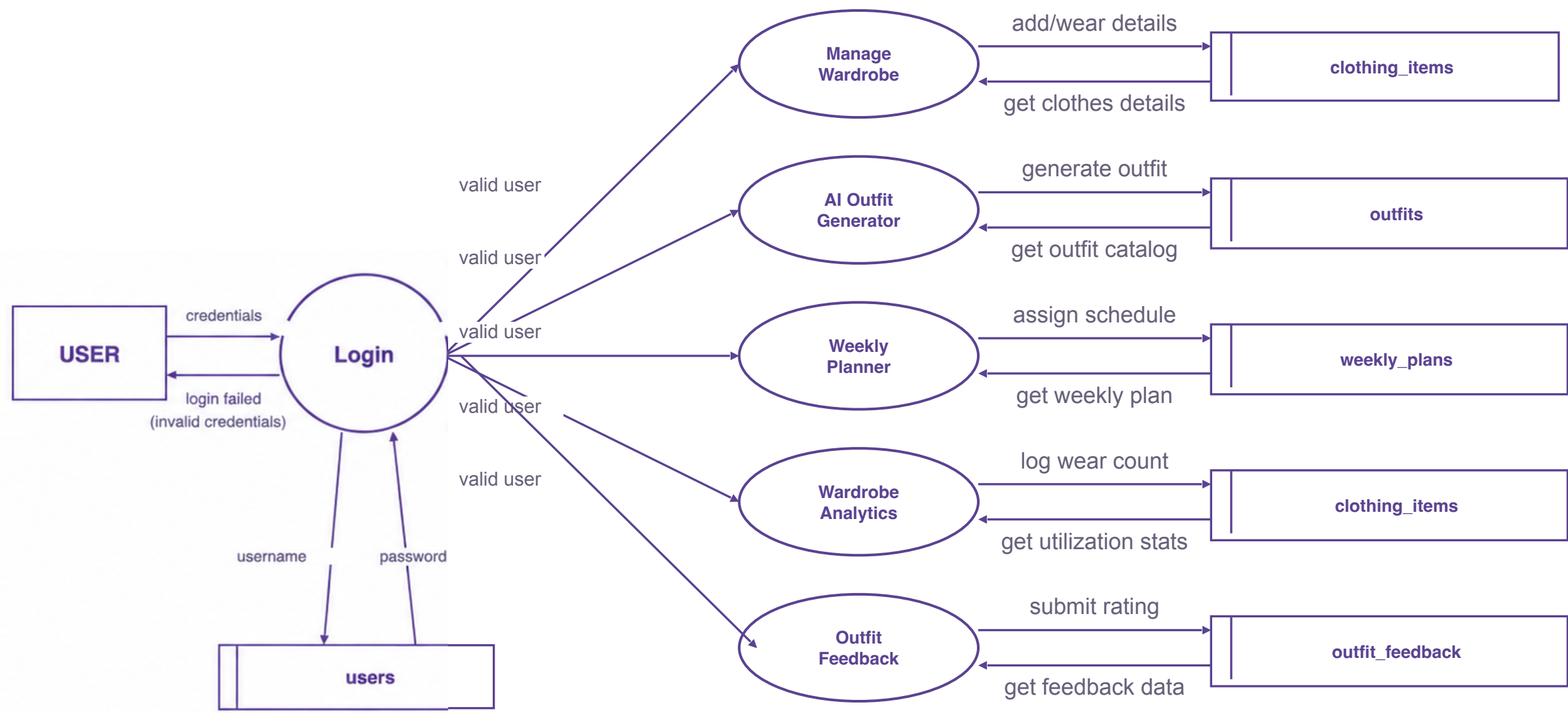


NOTATION KEY

-  External Entity
-  Process
-  Data Store
-  Labeled Data Flow

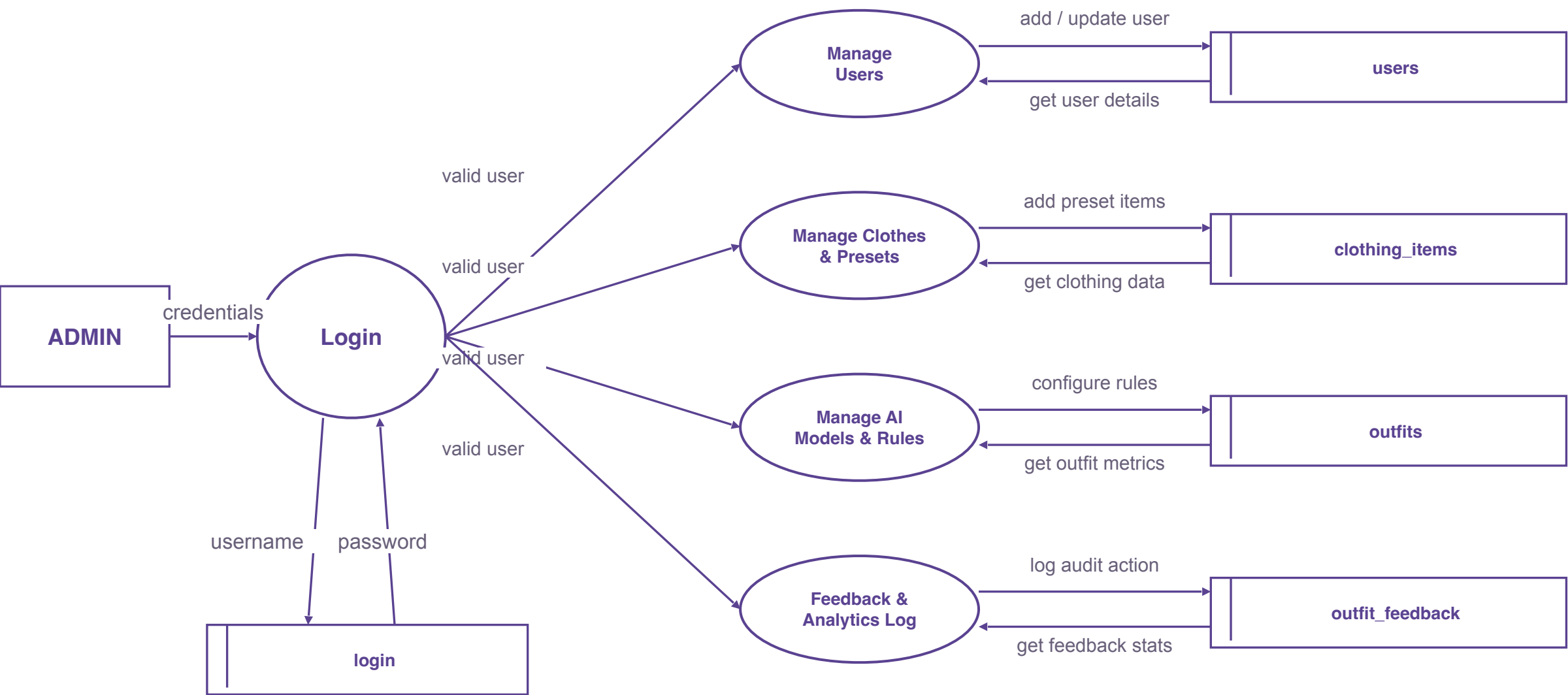
06 · SYSTEM DESIGN — DFD LEVEL 1

Decomposition of end-user functional flows in Wearlytics



06 · SYSTEM DESIGN — DFD LEVEL 1

Decomposition of administrator functional flows in Wearlytics



07 · DATASET & DATABASE

TRAINING DATASET

Clothing Dataset (Full) · Alexey Grigorev

Purpose: Train and evaluate MobileNetV2 for automatic clothing-cate

5,403

Annotated records

5,096

Usable records

6

Final categories

Source

Kaggle — “Clothing Dataset (Full)”

Key Features / Attributes

Image → 224 × 224 RGB tensor | Target → 6 clothing categories

Data Split

Training: 3,823 · Validation: 764 · Testing: 509

DATABASE

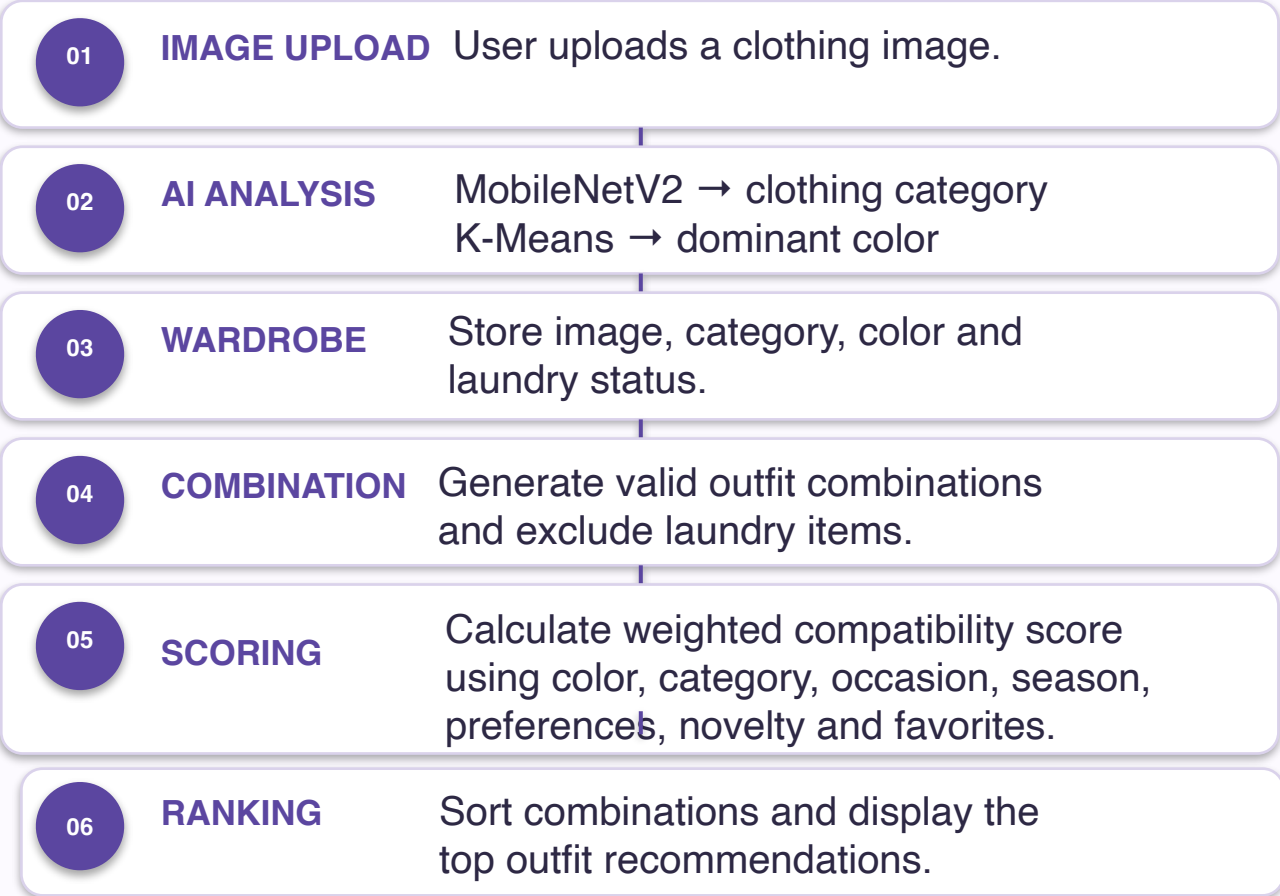
Wearlytics Application Schema

TABLE	PURPOSE
users	Accounts & authentication
clothing_items	Wardrobe items & AI metadata
outfits	Generated / saved outfits
outfit_items	Items belonging to outfits
weekly_plans	Seven-day outfit planning
user_preferences	User & learned preferences
outfit_feedback	Like / dislike feedback
custom_outfits	User-created custom outfits
custom_outfit_items	Items in custom outfits

Main fields: users (id, username, email) · clothing_items (id, user_id, type, color, image_path, season, occasion, wear_count, laundry)

Dataset → Clothing Classification |
Database → Wardrobe, Outfits, Planning, Feedback & Analytics

ALGORITHM WORKING



USER FEEDBACK & LEARNING Like / Dislike / Favorite → update learned preferences → improve future recommendations

CORE ALGORITHMS

- 1. MobileNetV2 — Clothing Classification
- 2. K-Means — Dominant Color Detection
- 3. Weighted Compatibility Scoring — Outfit Ranking

WEIGHTED COMPATIBILITY

Outfit score is calculated using:

Color Harmony	25%
Category Balance	20%
Occasion	20%
Season	15%
Preferences	12%
Novelty	5%
Favorites	3%
Total	100%

Higher score → higher recommendation rank

08 · PROPOSED METHOD / MODULES

Major Modules & Their Functions

Four integrated modules implemented in Wearlytics

01 USER & WARDROBE MANAGEMENT

- Registration & login
- Add, edit and delete clothing
- Upload images and manage metadata
- Laundry status & wardrobe availability

02 AI CLOTHING ANALYSIS

- MobileNetV2 clothing classification
- K-Means dominant color detection
- AI suggestions can be reviewed
- Category and color stored with item

03 AI OUTFIT RECOMMENDATION

- Generate valid outfit combinations
- Weighted compatibility scoring
- Color, category, occasion & season
- Preferences, novelty & favorites

04 PLANNING & ANALYTICS

- Save and manage outfits
- Seven-day weekly planner
- Wear tracking & last-worn information
- Utilization and wardrobe insights

WARDROBE → AI ANALYSIS → RECOMMENDATION → PLANNING & ANALYTICS

09 · RESULTS

Results & Model Evaluation

EXISTING SYSTEM

26.13%

OpenCV heuristic classifier

PROPOSED SYSTEM

89.19%

Fine-tuned MobileNetV2

ACCURACY IMPROVEMENT

+63.06 pp

absolute percentage-point gain

ACCURACY COMPARISON

System	Method	Correct / Total	Accuracy
Existing System	OpenCV rule-based heuristic (Aspect Ratio + HSV + Canny Edges)	133 / 509	26.13%
Proposed System	Fine-tuned MobileNetV2 (Transfer Learning)	454 / 509	89.19%
Improvement	Proposed – Existing		+63.06 pp

VALIDATION

91.88%

Best accuracy
Epoch 12 / 15

OTHER RELEVANT RESULTS

5,096

Dataset images

6 categories

509

Test images

10% held-out

454 / 509

Proposed correct

89.19% accuracy

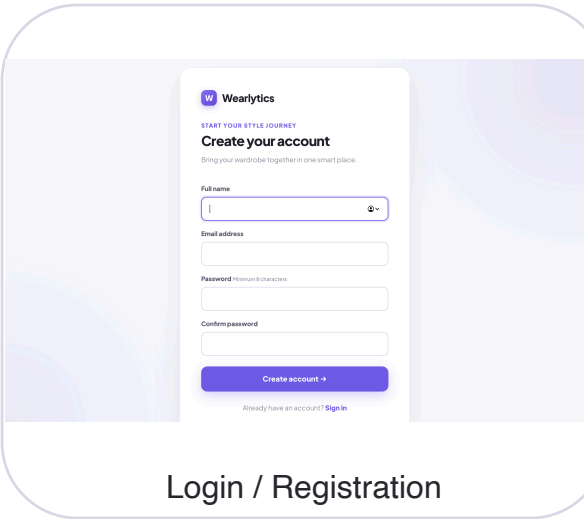
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Integration tests

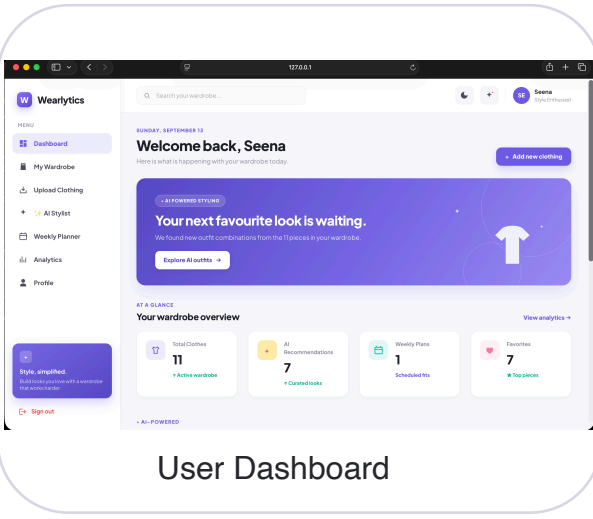
All passed

09 · RESULTS

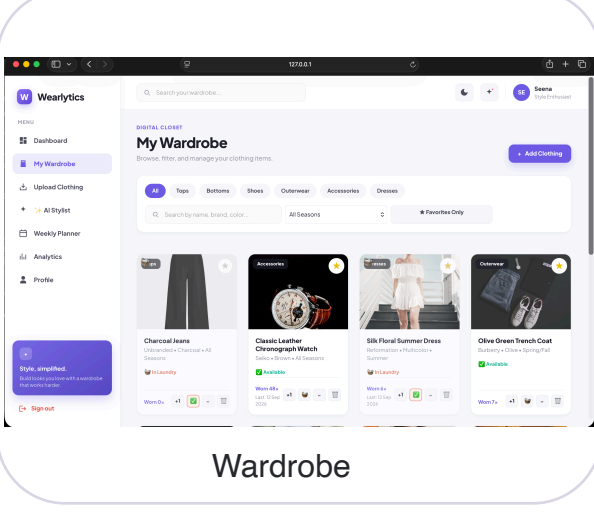
Implemented UI Screens



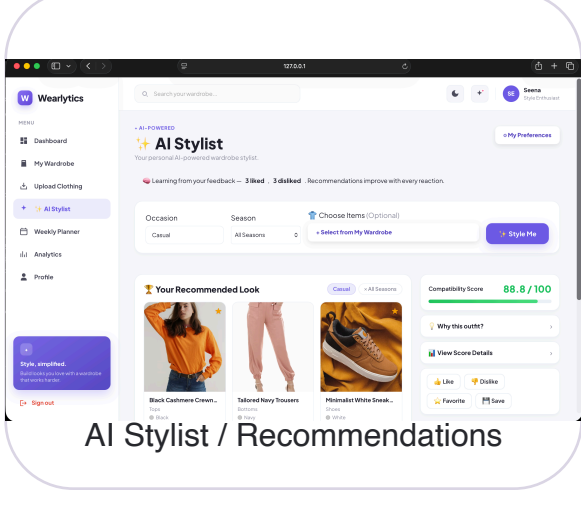
Login / Registration



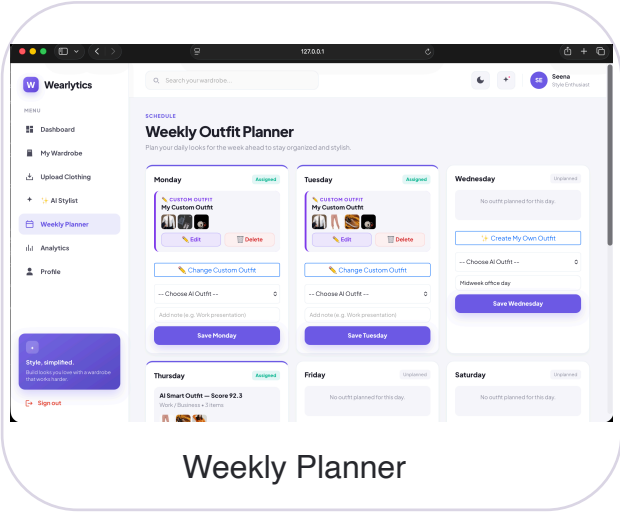
User Dashboard



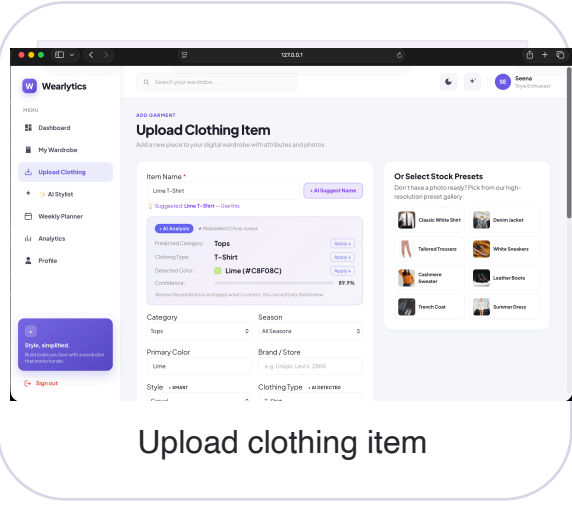
Wardrobe



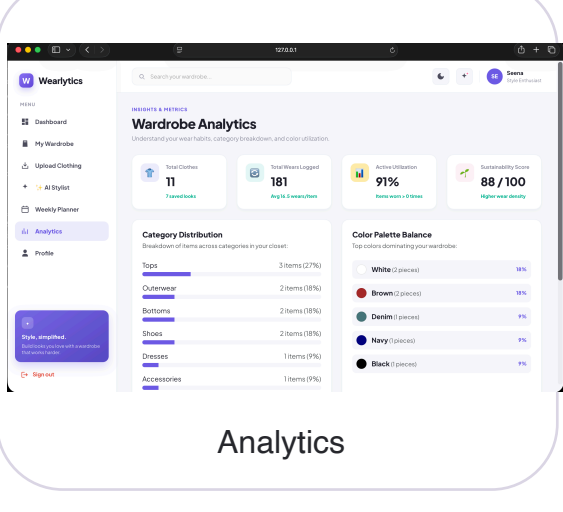
AI Stylist / Recommendations



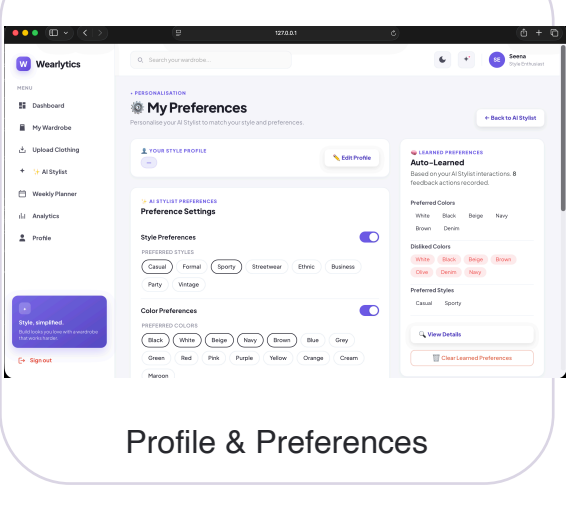
Weekly Planner



Upload clothing item



Analytics



Profile & Preferences

Conclusion & Future Enhancement

Completed work, achieved outcome and planned improvements

Conclusion

89.19%

Final test accuracy
509 test images

Delivered in this interim

- Digital wardrobe management
- AI clothing classification + color detection
- Weighted outfit recommendation
- Feedback-based preference learning
- Weekly planner + laundry awareness
- Wear tracking + wardrobe analytics

Achieved Outcome

A working smart-wardrobe system with measurable AI performance and transparent recommendation logic.

Future Enhancements

Possible improvements for the next version

- | | | | |
|-----------|---|-----------|---|
| 01 | Advanced recommendation models
Learned compatibility models and deeper personalization. | 02 | Dataset expansion
Larger, diverse and better-organized training data. |
| 03 | Full wear-event history
Store complete wear events and usage history. | 04 | Robust personalization
Make preference controls fully functional and reliable. |
| 05 | Classifier calibration
Improve confidence calibration and evaluation reliability. | 06 | Notifications
Laundry reminders and outfit/planner notifications. |
| 07 | Mobile application
Extend the system to native mobile access. | 08 | Production deployment
Deployment, scalability and stronger security/validation. |

Priority: stronger personalization · better evaluation · complete tracking · mobile & deployment

11 · REFERENCES

Key literature papers, dataset source and official technical documentation

RESEARCH PAPERS

01 Saed, S. & Teimourpour, B. (2026)

Saed, S. and Teimourpour, B., 2026. Hybrid-hierarchical fashion graph attention network for compatibility-oriented and personalized outfit recommendation. *Machine Learning with Applications*, p.100802.

02 Balloni, E., Pietrini, R., Frontoni, E., Mancini, A. & Paolanti, M. (2025)

Balloni, E., Pietrini, R., Frontoni, E., Mancini, A. and Paolanti, M., 2025. OutfitAI: shop the outfit with a deep learning-based intelligent expert system. *Multimedia Tools and Applications*, 84(32), pp.40195-40214.

03 Chang, C.-L., Chen, Y.-L. & Jiang, D.-X. (2025)

Chang, C.L., Chen, Y.L. and Jiang, D.X., 2025. Using large multimodal models to predict outfit compatibility. *Decision Support Systems*, 194, p.114457.

04 Suvarna, B. & Balakrishna, S. (2024)

Suvarna, B. and Balakrishna, S., 2024. *Enhanced content-based fashion recommendation system through deep ensemble classifier with transfer learning. Fash Text 11, 24 (2024)* [online]

05 Liu, J., Hou, L., Yu, X., Song, X. & Ren, Z. (2024)

Sun, K., Zhang, P., Zhang, J., Tao, J. and Yuan, K., 2024. PFNet: Attribute-aware personalized fashion editing with explainable fashion compatibility analysis. *Information Processing & Management*, 61(1), p.103540.

DATASET

Alexey Grigorev — Clothing Dataset (Full)

Kaggle

Used for clothing-image classification and model evaluation.

OFFICIAL TECHNICAL REFERENCES

- Flask Documentation
- PyTorch / torchvision Documentation
- SQLAlchemy Documentation
- scikit-learn Documentation
- Bootstrap Documentation

WEARLYTICS

Thank You

AI-Powered Smart Wardrobe and Personalized Outfit Recommendation System

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